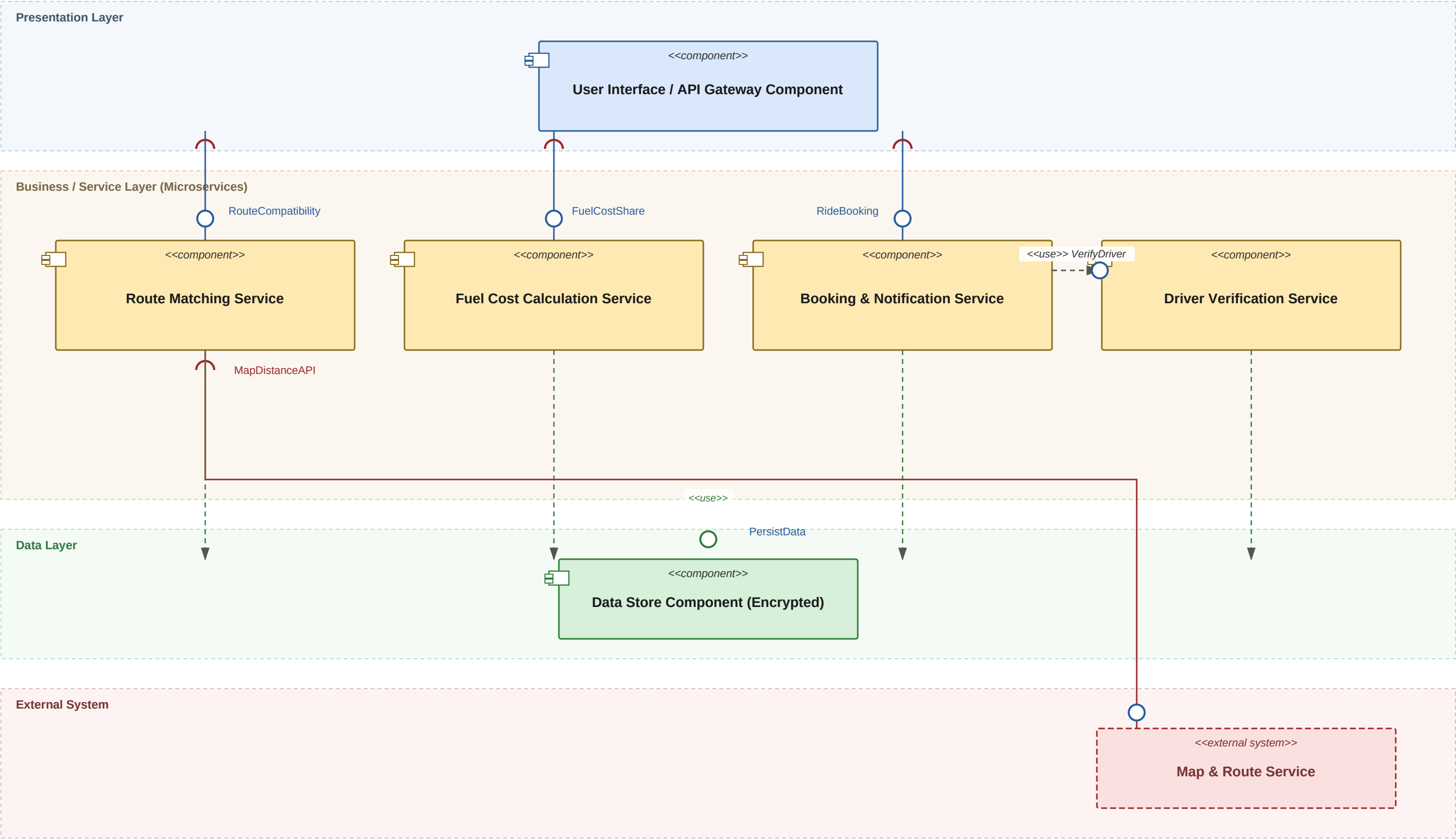


UML Component Diagram — Peer-to-Peer Carpooling & Expense Splitter (Microservices Architecture)



Lab 3: Component Modelling & Architectural Pattern Selection

Peer-to-Peer Carpooling & Expense Splitter

Architecture Selection: "We chose Microservices Architecture for the Peer-to-Peer Carpooling & Expense Splitter System."

Reason 1: Independent Scaling for Real-Time Route Matching

NFR-002 requires the system to resolve route-compatibility and detour-tolerance calculations within 2 seconds under up to 1,000 concurrent active queries, while other operations such as driver verification or notifications are far less frequent. A microservices architecture lets the Route Matching Service be deployed and scaled horizontally on its own — for example, running multiple replicas during peak commute hours — without over-provisioning the Fuel Cost Calculation, Booking, or Verification services, which see comparatively light and steady load.

Reason 2: Independent Deployment Across Distinct Functional Domains

The system spans functionally distinct domains — route compatibility matching (FR-001), automated fuel cost splitting (FR-002), recurring schedule matching (FR-003), driver document verification (FR-004), and the ride request/approval workflow (FR-005). Each domain evolves on its own release cadence (e.g., fuel-price logic changes more often than verification rules) and can fail independently. Microservices allow each capability to be built, tested, deployed, and rolled back in isolation, so a defect or outage in one service (e.g., Fuel Cost Calculation) does not take down core booking functionality.

Security Advantage

NFR-001 requires driving-license verification documents and travel histories to be encrypted at rest (AES-256) and in transit (TLS 1.3). Isolating the Driver Verification Service and the encrypted Data Store Component as separate, independently secured components restricts access to sensitive PII and legal documents to a single, narrowly scoped service boundary — rather than exposing that data across the whole application — reducing the attack surface and simplifying compliance auditing.

Performance Benefit

Because the Route Matching Service is decoupled from the rest of the system, it can be independently optimized and horizontally scaled (e.g., additional compute instances, geospatial indexing, caching of the Map & Route Service responses) to consistently meet the 2-second / 1,000-concurrent-query target in NFR-002, without requiring the entire monolith to be scaled up at proportionally higher cost.